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PlanetX. Manual

Русская версия

Version 0.16.0

PlanetX is a 3D globe inside QGIS in the spirit of Google Earth. The globe opens in its own window and shows the whole Earth with terrain and atmosphere, from space down to single streets. Layers of the current project, your own placemarks, paths and polygons lie on the globe. Tours fly over the places, and a tour recording gives frames for a video.

Plugin page - www.informpp.ru.

Installation

The plugin runs in QGIS 3.36 and newer, including QGIS 4. It is tested on QGIS 3.36 with Qt 5 and on QGIS 4.0 with Qt 6. It needs a graphics card with OpenGL 3.3 and the Python module PyOpenGL. QGIS builds for Windows include PyOpenGL.

Open **Plugins** → **Manage and Install Plugins**, search for PlanetX and install it. QGIS does not need a restart.

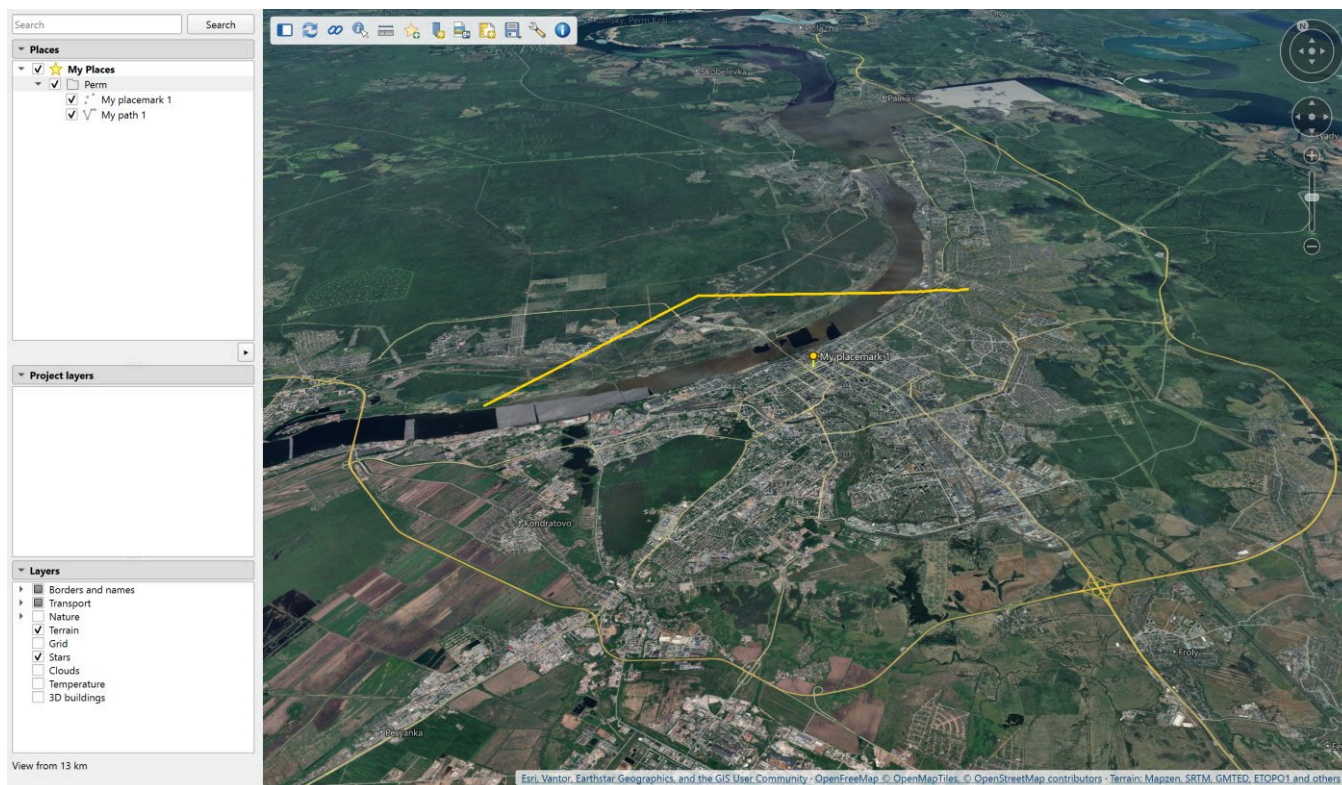
The globe opens with the button on the PlanetX toolbar or from **Web** → **PlanetX**. The same menu holds About PlanetX. If the QGIS build has no PyOpenGL, the plugin says so and does not open the window.

Interface language

The language follows the QGIS settings, not the system. With Russian in QGIS the plugin is in Russian, with any other language it is in English. The label language on the globe is set separately, in the view properties.

The globe window

The panel is on the left, the globe view on the right. The icon bar sits in the top left corner of the view, the data source credits in the bottom right corner. The border between the panel and the view is dragged with the mouse.



Icon bar

Icon	What it does
Sidebar	Hides and shows the left panel, the view takes its place
Refresh	Shows new settings and layers. It turns orange when there is something to show
Synchronization	Links the globe with the QGIS map window
Identify features	Turns on identify by a click on the globe
Ruler	Opens the Ruler window
New placemark	Opens the New placemark window
Save view	Puts a placemark at the look-at point with height, heading and tilt
View snapshot	Saves the view to a PNG or JPEG file
View to layout	Puts the view into a QGIS layout as a picture
Scene	The menu Save Scene... and Open Scene...
View properties	Base map, terrain, labels, link with the map, update, navigation controls
About	Controls, data sources, links

Status line

The status line sits under the left panel. It usually shows the camera height, for example "Overview from 2 000 km". The second line shows the coordinates and elevation of the point under the cursor.

The line also reports loading:

- "loading 12" - this many tiles, layer pictures and labels still wait for loading.
- "loading has stalled for 7 s" - data waits and nothing has arrived for 5 seconds or longer.

- "places are heavy, 250 thousand vertices" - your own objects exceed 200 thousand vertices, turning may become slow.
- "The base map did not load" and "The vector base did not load" - the source answered with an error.

While loading goes on, a blue icon spins to the right of the icon bar. A short load does not light it, it appears after half a second. When loading stalls, the icon turns orange. Its tooltip tells how much still waits.

Messages of search, scenes, tours and recording stay for 5 seconds.

Navigation

Navigation follows Google Earth.

Mouse

Action	What happens
Left button drag	The Earth follows the cursor. After release it spins on by inertia
Wheel	Zooms in and out. The point under the cursor stays in place
Left double click	Flies to the point under the cursor and zooms in 2.5 times
Right double click	Zooms out 2.5 times
Right button drag	Up - closer, down - farther, toward the press point
Middle button or left with Shift	Turn and tilt around the look-at point
Left button with Ctrl	Looks around, the eye stays in place

The camera does not go lower than 50 m above the terrain. The tilt is limited to 85°. While the ruler, drawing or identify is on, a double click stays two plain clicks and places points.

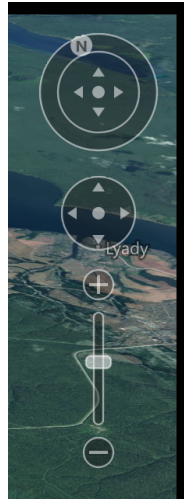
Keys

Keys work when the globe view has focus. A click on the globe gives it focus.

Key	What happens
Arrows	Move the view by a twentieth of the window width
Shift and arrows	Turn and tilt by 3°
Ctrl and arrows	Look around by 3°
PageUp, plus	Zoom in by a wheel step
PageDown, minus	Zoom out by a wheel step
N	North up
U	Look straight down
R	North up and look straight down
Space	Stop inertia and flight

On-screen controls

The navigation controls stand in the top right corner of the view, as in Google Earth.



Control	What it does
Compass ring	Dragging turns the view. The letter N shows north
Letter N	A click puts north up
Stick inside the ring	While the button is held, the view looks toward the press
Lower stick	While the button is held, the view moves toward the press
Plus and minus	While the button is held, the camera zooms in or out smoothly
Slider	The thumb stands at the camera height, dragging sets a new one

The speed of the sticks grows with the distance from the middle. The controls appear when the cursor comes near the corner of the view. The check box Navigation controls always on screen in the view properties keeps them all the time.

Place search

Type a place name into the Search field at the top of the panel, for example `Perm`, or coordinates in degrees, for example `58.0105, 56.2294`. Enter or the Search button starts the search.

- Coordinates start a flight at once. The camera lands no higher than 2 km from the point.
- A name goes to the Nominatim service in the label language. The camera flies to the first place found and frames it whole.
- When several places are found, they are listed under the field. A click on a row starts a flight.
- The found place carries a red pin. Clearing the field removes the pin and closes the list.

Over a long distance the camera rises and lands smoothly. The mouse interrupts a flight.

Left panel

The panel has three sections - Places, Project layers and Layers. A click on a header collapses the section, and the open sections take its place. The globe remembers which sections are collapsed.

Layers section

The section follows the Layers panel of Google Earth. Its check boxes take effect at once, without the Refresh button.

Group	Rows
Borders and names	Borders, Places, Water names
Transport	Roads, Road numbers, Railways, Airports
Nature	Rivers, Lakes and reservoirs, Peaks, Reserves and national parks
Terrain	Mapzen Terrain Tiles elevations and hill shading
Grid	Parallels and meridians with labels, the equator, tropics and polar circles in yellow
Stars	Stars brighter than magnitude 6 and the Milky Way at their places in the sky
Clouds	Clouds from NASA GIBS VIIRS imagery of the last complete day
Temperature	Surface temperature of land by day over 8 days (MODIS) and of the sea over a day (GHR SST MUR) with a scale in degrees
3D buildings	OpenStreetMap buildings as blocks from OpenFreeMap tiles, off by default

The vector base comes from OpenFreeMap tiles. On the first opening borders, places, terrain and stars are on. The tooltip of each row tells how the feature is drawn and from what height it is visible.

The grid step follows the camera height, from 30° from space to seconds near the ground. The stars stand where they stand over the Earth at this minute and fade when the camera goes below 150 km. The Milky Way image, about 10 MB, is downloaded when the stars are first shown and comes from the QGIS cache afterwards. Clouds lie as a veil over the imagery. Snow and ice get into the clouds too, they have the same colour.

Temperature colours land and sea with their own scales, the scales in degrees Celsius are in the bottom left corner of the view. Land shows the temperature of the surface itself by day, not of the air, over 8 days. Land may have gaps under clouds. Sea shows the water temperature near the surface over a day.

Labels stay level at any turn and tilt and do not overlap. A place behind a mountain or beyond the horizon has no label.

3D buildings

The 3D buildings row of the Layers section puts OpenStreetMap buildings on the globe as blocks. Outlines and heights come from the same OpenFreeMap vector tiles as the vector base. The row is off by default.



Buildings show when the camera is closer than 6 km to the ground. The globe keeps up to 32 building tiles around the camera, the nearest first. Tiles behind the camera are not loaded.

The height of a building is its height in OpenStreetMap, otherwise the number of floors times 3.66 m. A building without either gets a height of 5 m. In the centre of Perm two thirds of the buildings have a height of their own, in the centre of Moscow almost all. In the outskirts many houses have the same height. The colour of a building comes from OpenStreetMap, otherwise the building is light grey. Roofs are flat.

The bottom of a building stands at the lowest point of the terrain under its outline. A view snapshot waits until all buildings of the frame are loaded. A scene remembers whether buildings are on.

View properties

Base map

Esri World Imagery - example

Terrain

Vertical exaggeration1.0

Labels

LanguageAs in QGIS

QGIS map

SynchronizationBoth ways

☐ Layers as on the QGIS map

☐ New layers straight to the globe

Update

☐ Update automatically

Navigation

☐ Navigation controls always on screen

Закрыць

The View properties icon on the icon bar opens the View properties window. The window does not block work with the globe.

Group	Field	What it sets
Base map	List	Source of imagery or map
Terrain	Vertical exaggeration	Height multiplier from 0.5 to 10
Labels	Language	Language of labels and search
QGIS map	Synchronization	Direction of the link with the map
QGIS map	Layers as on the QGIS map	The globe shows the layers switched on in the QGIS legend
QGIS map	New layers straight to the globe	A new project layer is checked on the globe at once
Update	Update automatically	The globe refreshes after every change without the Refresh button
Navigation	Navigation controls always on screen	The ring, sticks and slider stay all the time, not only near the corner

The default base map is Esri World Imagery, listed as an example. Esri sets its terms of use. OpenStreetMap comes next, then the XYZ Tiles connections from the QGIS browser. Terrain connections are not listed. A new connection appears in the list the next time the window opens.

The label language is chosen from a list. As in QGIS takes the QGIS interface language, Local names keep the names in the language of the country. Fifteen languages follow. When there is no name in the chosen language, Latin-script languages get the Latin name, the others the local one.

A change of base map, terrain exaggeration or project layers shows after the Refresh button on the icon bar. The status line reminds about it. The label language changes at once.

Project layers section

The section lists the layers of the current project in the order of the QGIS map, each with its type - raster, points, lines, polygons or table. The check box of a layer shows it on the globe and does not change its visibility on the map. In a new project the boxes are clear. The checks are stored in the project.

Layers lie on the globe as a picture drawn by QGIS, with the styles of the map. The picture lies on the terrain, it cannot be raised or extruded.

Labels of vector layers come separately, like city names. They stay level at any turn and tilt, do not overlap and hide behind mountains and the horizon. The text, colour and size come from the layer labels in QGIS, rule-based labels too. A point label stands to the right of the point, a line label at its middle, a polygon label inside it. Features around the view point are labelled, at most 500 per layer. The scale visibility of QGIS labels is not used on the globe.

A double click on a layer flies to its extent, looking straight down. The right-click menu:

- Fly to - a flight to the extent of the layer.
- Transparency slider - the transparency of the QGIS layer itself, it changes on the map too.
- Track... - for point layers, see [Tracks](#).
- Layer Properties... - the standard QGIS layer properties window.

Places section

The section holds the My Places folder, described in the chapter [My Places](#).

Link with the QGIS map

Synchronization

The synchronization icon on the icon bar links the globe with the QGIS map window. The direction is chosen in the view properties:

- Both ways - the default.
- Map leads the globe - the globe moves to the map extent and keeps its tilt and heading.
- Globe leads the map - the map centers on the look-at point of the globe once the globe stops.

Any map coordinate system works. The map width on the ground equals the width of the globe view.

Selection

Features selected on the QGIS map are highlighted on the globe.

Identify features

The Identify features icon turns identify on, the cursor becomes a cross. A click on the globe puts a pin with the coordinates and opens the Features window.

- The top of the window shows the coordinates and elevation of the point.
 - Below are the project layers checked on the globe, their features under the point and the feature fields. A raster shows its values at the point.
 - The Select on map button selects the found features in the QGIS layers. The selection shows on the map, in the attribute table and on the globe.
-

My Places

My Places is the folder at the top of the Places section, marked with a star. It holds placemarks, paths, polygons, saved views and measurements. The places are kept in one file `PlanetX/myp1aces.gpkg` in the QGIS profile folder and are visible in any project, like My Places in Google Earth.

List rows

The row icon shows the kind of object - placemark, path, polygon, saved view or folder. A check box shows and hides the object on the globe, a folder check box does it for the whole folder. A double click on a place flies there.

Places and folders are rearranged by dragging, including from folder to folder. The tour follows the list order.

Several rows are selected with Ctrl and Shift. The selection is dragged together, and the Del key deletes it after a question.

Menus

Where	Items
My Places	Play tour, New Folder, Open KML or KMZ..., Save as KML..., Copy, Paste, Add the places layers to the project
Folder	Play tour, New Folder, Open KML or KMZ..., Save as KML..., Copy, Paste, Rename..., Delete
Place	Fly to, Tour along the path, Snapshot view, Properties..., New Folder, Copy, Paste, Rename..., Delete
Several rows	Copy, Show selected, Hide selected, Delete selected
Empty space	New Folder, Open KML or KMZ..., Paste

Only paths have Tour along the path.

A new folder appears at once with the name New Folder and is renamed later. On a folder it is created inside, on a place right below it. A folder is deleted with its contents after a question.

Add the places layers to the project adds the three layers of the places file to the project - points, lines and polygons. Their attributes and undo are the standard ones.

New placemark

The New placemark icon opens a window with the tabs Placemark, Path and Polygon.

- A placemark is put with a click on the globe, a new click moves it.
- Path points and polygon vertices are put with clicks on the globe.
- The Name field is filled at once with a numbered name - My placemark 1, My path 1, My polygon 1. Each kind counts on its own. The name can be changed in the same field.
- Color sets the color of the line and outline, the polygon fill has the same color, half transparent. Width is set in screen pixels.

The Save button writes the object to the selected folder of My Places. The window stays open for the next object. The Clear button removes the points put so far.

Saved view

The Save view icon puts a placemark at the look-at point. The name is offered with a number - View 1, View 2. The placemark keeps the camera distance, heading and tilt. A flight to such a placemark brings the whole view back.

Place properties

Name

My path 1

Description

Color

Width

3.00

Height above ground

0.0 m

Ground

Space

Extend to ground

Place view

Latitude

58.035000°

Longitude

56.210000°

Range

3000 m

Heading

0.0°

Tilt

0.0°

Snapshot current view

Reset

OK

Отмена

The Properties... item opens the properties window. The window does not block the globe, the view can be turned and zoomed. Each change shows on the globe at once. OK writes the changes to the place, Cancel brings the previous look back.

Field	What it sets
Name	Name in the list and label on the globe
Description	Text of the placemark, it goes into KML too
Color, Width	Color and width of the line and outline
Fill	Fill color of the polygon and of the wall
Height above ground	Lift of the object above the terrain in metres. The Ground - Space slider under the field sets it from the ground to 100 km
Extend to ground	A wall from the object to the ground, a post for a placemark
Place view	Look point, range, heading and tilt of the camera at the place

Extending works with a height above zero. A path becomes a wall, a polygon becomes a block.

Place view

Any place can have a view of its own, like the View tab in Google Earth. A view is the point the camera looks at, the range to it, the heading and the tilt. The look point may differ from the place. Fly to, a double click on the place and a tour stop follow the view.

Snapshot view in the place menu makes the current globe view the view of the place. The properties window has the same values in the Place view section. The Snapshot current view button takes the globe view, Reset brings back the view the place had when the window opened. Clearing the section check box removes the

view, then the camera frames the whole place. The view goes into KML as a LookAt element and is read from it.

Copy and paste

Copy or Ctrl+C puts the selected places and folders into the clipboard as KML text, as Google Earth does. The text can be pasted into a text editor, edited and copied back. Paste or Ctrl+V puts the places from the clipboard into a folder. On a place it is the folder of that place, on empty space the root of My Places. Folders from the clipboard come in as folders. KML copied in Google Earth pastes the same way.

KML and KMZ

Open KML or KMZ... puts a Google Earth file into My Places as a new folder named after the file. Folders, styles and placemark views are kept. The camera flies to the contents of the file.

Save as KML... saves a folder or the whole My Places to KMZ or KML. The file extension chooses the format.

Ruler

The Ruler icon opens a window with the tabs Line, Path, Polygon and Circle. Points are put with clicks on the globe, a rubber band follows the cursor.

Tab	Values
Line	Length. The third click starts a new line
Path	Length
Polygon	Perimeter, area
Circle	Radius, perimeter, area. The first click is the center, the second sets the radius

The values are computed on the WGS84 ellipsoid. Each value has its own units - metres, kilometres, miles, nautical miles, and for the area square metres, hectares, square kilometres, square miles.

The Save button puts the shape into My Places together with the measurement. The name is offered with a number, for example Line 1. The measurement shows in the row tooltip.

The Ruler and New placemark windows are open one at a time. Opening one closes the other.

Tours

A tour flies over the checked places of a folder in the list order, including nested folders.

- A place with a view of its own brings back its look point, range, heading and tilt.
- Along a path the camera travels the line.
- Other places are framed whole.

A tour starts with Play tour in the menu of My Places or a folder. A path has Tour along the path in its menu. The ► button under the places list starts a tour too. When no places are checked, the status line says so.

The tour bar appears at the bottom of the view:

Button	What it does
Slider	Shows how much of the tour has passed and winds the tour. The camera moves to the chosen point at once, after release the tour goes on from it

Button	What it does
⏮, ⏭	Previous and next stop
⏸, ▶	Pause and continue
Pause	How many seconds the camera stays at a stop
●	Records the tour as PNG frames
×	Ends the tour

Moving the camera with the mouse pauses the tour.

Recording a tour for a video

The ● button asks for a folder and records the tour into it as PNG frames, 25 frames per second of the tour. The frame size equals the window size. The source credits stand in the corner of each frame. The files are named `frame_00000.png`, `frame_00001.png` and on, frames with the same numbers in the folder are replaced. Each frame waits for the tiles, so recording takes longer than the tour and the frames have no blurred spots. A frame that waits longer than 20 seconds is taken as it is. Recording always starts at the first stop, so a repeated recording gives the same frames.

A file `frames.json` lies next to the frames - frame rate, size, camera pose and data time of each frame, numbers of frames that did not fully load.

The bar shows the frame number. The cross on the bar stops the recording.

Making a video from the frames

PlanetX writes frames, not a video file. Any program that opens an image sequence assembles a video from them.

The free program ffmpeg (ffmpeg.org) makes an MP4 with one command in the folder with the frames:

```
ffmpeg -framerate 25 -i frame_%05d.png -vf "pad=ceil(iw/2)*2:ceil(ih/2)*2" -c:v libx264 -pix_fmt yuv420p -crf 18 tour.mp4
```

The rate 25 matches the recording rate, so the video runs at the pace of the tour. The `pad` filter adds a pixel when a frame side is odd, the H.264 codec needs even sides. A smaller `-crf` value gives better quality and a larger file.

The video editors Shotcut and DaVinci Resolve open numbered frames as one clip. Music, captions and several tours joined together can be added there.

Tracks

A track shows the path of moving objects over time. The source is a point layer of the project with a time field, and the QGIS Temporal Controller sets the time.

The Track... item in the menu of a point layer opens the settings window:

Field	What it sets
Time	A field with date and time, ISO 8601 text or seconds since 1970
Object	A field that tells the objects apart. One object - all points form one path
Color	Color of the travelled path

Field	What it sets
Camera follows	The camera keeps the first object in the middle of the view along the motion

Each object grows its travelled path up to the end of the current controller range, and a label with its name marks the current point. With the controller off, whole tracks are shown. The Remove track button returns the layer to its usual look. Track settings are stored in the project.

During a tour recording the tracks grow along the tour over the whole controller range.

Scenes

A scene saves the whole view and can be passed to other people. The Scene icon opens the menu Save Scene... and Open Scene.... A scene file has the extension `.planetx`.

A scene holds:

- the camera pose.
- the time of the Temporal Controller.
- the project layers checked on the globe, as a link to their source.
- the base map, terrain, terrain exaggeration, groups of the Layers section and the label language.
- the rows of the Layers section, the grid, stars, clouds, temperature and 3D buildings.
- the selected folder of My Places with its tour.

A database password does not go into the file, a link to a QGIS connection setting stays. When the whole My Places folder or nothing is selected, no places go into the scene.

When a scene opens, a layer is looked up in the project by its source and added if missing. The other project layers are unchecked on the globe. The places of the scene come in as a new folder, and the camera flies to the scene pose. The status line lists the layers that were not found and did not open.

View snapshot and view to layout

View snapshot

The View snapshot icon opens a window with the snapshot width and height in pixels. A snapshot can be larger than the window, the globe loads detailed tiles for it. The Window proportions check box keeps the height in step with the width.

The Save... button asks for a PNG or JPEG file and starts the snapshot. The camera stands still meanwhile. The Take now button does not wait for the remaining tiles. The source credits stand in the bottom right corner of the snapshot.

View to layout

The View to layout icon puts the view as a fixed picture into a QGIS layout, an existing one or a new one.

Field	What it sets
Layout	An existing layout or New layout
Width, Height	Size of the picture on the page in millimetres
Resolution	Output resolution of the layout, it sets the snapshot size in pixels

The picture is kept inside the project. The layout window opens after the insertion.

Limitations

- Project layers lie on the terrain as a picture, a layer cannot be raised or extruded by a field. Your own places have lift and extrusion.
- Pictures of project layers are not filtered by the controller time.
- Above latitude 85° the poles are covered with the ocean color, the Web Mercator tile grid ends there.
- Recording a tour takes longer than the tour itself.
- Buildings are blocks with flat roofs, roof shapes and facades are not shown.

Data sources

Imagery - Esri World Imagery, Esri sets the terms. Map data © OpenStreetMap contributors. Vector tiles and 3D buildings - OpenFreeMap. Terrain - Mapzen Terrain Tiles, SRTM, GMTED, ETOPO1 and other data. Clouds - NASA GIBS, VIIRS imagery. Temperature - NASA GIBS, MODIS and GHRST MUR. Stars - the Yale Bright Star Catalogue. Milky Way - NASA/Goddard Space Flight Center Scientific Visualization Studio, Gaia DR2: ESA/Gaia/DPAC. Place search - Nominatim.

The terms of all sources are described in [SOURCES.md](#). Tiles go through the QGIS network settings and cache, the plugin keeps no cache of its own.

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